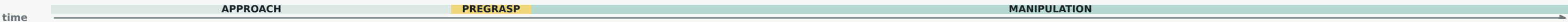


ONE SCENE, TWO LATENT WORLDS

HELD-OUT LIBERO · PAIRED PROMPTS

Counterfactual prompts remap the same visual state into goal-centric representations that remain readable inside action computation.

A One physical trajectory; two counterfactual goals



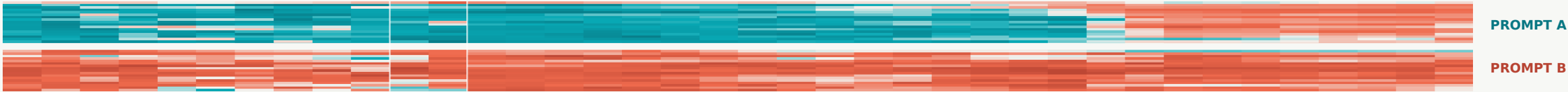
IDENTICAL RGB + ROBOT STATE —————> PROMPT A Pick up Alphabet Soup —————> PROMPT B Pick up Cream Cheese

B Layer-by-time tomography of decoded target identity

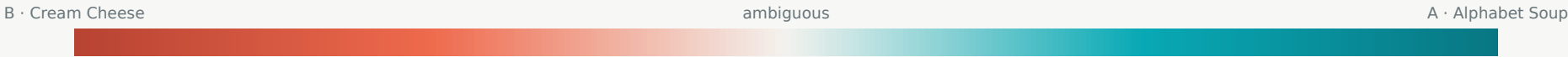
OpenVLA
visual-language policy · 32 blocks



pi0.5 VLM
PaliGemma prefix · 18 blocks



pi0.5 Expert
action-token mean · 18 blocks



C At one pregrasp state, decoded goal coordinates bifurcate across depth

